

Random matrix theory analysis of cross-correlation in the Nigerian Stock Exchange

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Abstract

In this paper we use Random Matrix Theory (RMT) to analyse the eigenstructure of the empirical correlations of 82 stocks which are consistently traded in the Nigerian Stock Exchange (NSE) over a 4-year study period 3 August 2009 to 26 August 2013. We apply the Marcenko-Pastur distribution of eigenvalues of a purely random matrix to investigate the presence of investment-pertinent information contained in the empirical correlation matrix of the selected stocks. We use hypothesised standard normal distribution of eigenvector components from RMT to assess deviations of the empirical eigenvectors to this distribution for different eigenvalues. We also use the Inverse Participation Ratio to measure the deviation of eigenvectors of the empirical correlation matrix from RMT results. These preliminary results on the dynamics of asset price correlations in the NSE are important for improving risk-return trade-offs associated with Markowitz's portfolio optimization in the stock exchange, which we achieve by cleaning up the correlation matrix. We propose to measure extent of closeness or otherwise in selected sectors of the NSE and the Johannesburg Stock Exchange (JSE) in our future work.

Key terms: Random matrix theory; Correlation matrix; Inverse participation ratio; Eigenvalue and eigenvector; Portfolio optimization.

1. Introduction

Estimation of risk and expected returns based on the standard deviation and expected value of returns was the main work of Markowitz's model in (1952). Since then, researchers have been working to improve the model and develop other models to manage and optimize portfolios. Random matrix theory (RMT) which was first proposed and introduced by (Wigner 1951) to explain statistical fluctuations of neutron resonances in the compound nucleus, became a popular technical tool for investigating the dynamics of financial markets using cross-correlations of selected financial data in the markets. (Wigner 1955) describes the general properties of the energy levels of highly excited states of heavy nuclei as measured in nuclear reactions, (Izenman 2008). He assumes that the interactions between the constituents of the nucleus are so complex that they should be modelled as random fluctuations. He represents it by Hermitian operator H (called Hamiltonian), which behaves like a large random matrix. The energy level of the system was approximated by the eigenvalues of this matrix, and the

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spacing between the energy levels of the nuclei could be modelled by the spacing of the eigenvalues of the matrix.

Analogically, the complex interrelationships and correlations among different assets in a globalised financial market, with open access to assets and information across markets, can be modelled against a baseline of random correlation matrix behaviour which represents complete unpredictability of a market. Hence, investors will be able to construct more optimal portfolios if they can locate within the correlation structure of a large number of assets those features that identify significant or pertinent information, compared to a null hypothesis purely random correlation matrix. While these lines of RMT work have been attempted in many markets (see examples below), there are no studies known to the authors which use such techniques in exploring the dynamics of correlation matrices among stocks from the Nigerian Stock Exchange. This paper presents related preliminary results.

Researchers like (Dyson 1962), (Dyson and Mehta 1963) and (Mehta 1991) work on RMT, which is a vestige of techniques used in mathematical physics, (Mehta 2004). (Laloux et al. 1999) and (Plerou et al. 1999) introduce RMT in financial markets where it is used in studying the statistical properties of cross-correlations of financial variables in the markets. (Sharifi et al. 2003) apply RMT to an empirically measured financial correlation matrix and show that the matrix contains a large amount of noise. They also attempt to separate the noisy part from the non-noisy part of the matrix, a process that is fondly referred to as cleaning the correlation matrix (Alaoui, 2015).

(Rak et al. 2007, 2008) apply RMT in the Warsaw stock market, by using the correlation matrix approach to study the temporal evolution of the Warsaw Stock Market as represented by the WIG20 index in 2006. Their 2007 paper investigates the inter-stock correlation for the largest companies listed on Warsaw Stock Exchange and WIG20 index. Their results infer that Warsaw Stock Market can be well described by a one factor model. A useful insight from this focus on large stocks is that RMT can be used to systematically characterise a stock market by a) using as many stocks as possible to model inter-stock correlations at an overall market level, and b) performing similar analyses for distinct sectors of the market, for example banking and financial services, agriculture, conglomerates, consumer goods, ICT, healthcare, industrial goods, oil and gas. This paper addresses the overall market perspective and informs subsequent work on sector-based systematic stock market characterisation and development (Ezepue and Omar, 2012).

A related insight is the need to run comparative RMT-based characterisation analyses across different epochs likely to correspond with policy in economic policy, structural macroeconomic and market structure environments in a financial system. Examples of these epochal events are the 2004 bank reforms in Nigeria, the 2007-09 global financial crisis and the recent UK Brexit by which UK will leave the European Union.

For an overview of RMT applications in different contexts, (Wilcox and Gebbie 2007) construct correlation matrix from 10 years daily data obtained from the Johannesburg Stock Exchange, and apply RMT to compare correlation matrix estimator obtained from the market with similar-sized random matrix. (Kulkarni and Deo 2007) and (Pan and Sinha 2007) work on the Indian stock market, while (Curkur et al. 2007) analyse the Istanbul stock market. (Wang et al. 2013) examine the statistical properties of cross-correlations in the US market and note that the detrended cross-correlation analysis (DCCA) coefficient method has similar

results and properties with Pearson's correlation coefficient (PCC). (El Alaoui 2015) study cross-correlations among stocks of Casablanca Stock Exchange using RMT. They tried to observe if the difference between predicted risk and realized risk will be reduced by cleaning the noisy element of the correlation matrix.

In this paper, we use the methods of (El Alaoui 2015) to perform the overall market characterisation analysis foreshadowed above. We also analyse the eigenvectors through their distributions and computation of the inverse participation ratio, a statistical measure of the inherent dynamics of the stock market linked to the cross-correlations of stock returns; see further notes below. We also explain how the information from RMT eigenstructure analysis that could be used to clean the correlation matrix of Nigerian Stock Exchange in order to reduce the gap between predicted risk and realized risks reflected in Markowitz's mean-variance Efficient Frontiers.

The rest of the paper is organised as follows. In section 2, we describe the empirical data used. Section 3 presents the methodologies and theoretical background on Random Matrix Theory. Section 4 shows and discusses the main empirical results and Section 5 concludes the paper.

2. Data

The Nigerian stock Exchange (NSE) has 188 stocks listed under eleven (11) sectors, namely: Agriculture with 5 stocks, Conglomerates with 6 stocks, Construction/Real estate with 9 stocks, Consumer goods with 28 stocks, Financial services with 57 stocks, ICT with 9 stocks, Health care with 11, Industrial goods with 21, Natural resources, Oil and Gas, and Services with 5, 14 and 23 stocks, respectively.

The data set used is the daily closing price of 4 years stock data listed in the NSE. We have 1019 daily closing prices running from 3rd August 2009 to 26th August 2013, excluding weekends and public holidays in Nigeria (Nationwide). These stock price data were converted into 1018 logarithmic returns. These data were screened to remove stocks that were delisted, infrequently traded or not traded at all during the study period. This reduced the data to 82 securities only.

Let $P_i(t)$ be the closing price of the index on day (t) of stock i and define the natural logarithmic returns of the index (i.e. the log-difference of $P_i(t + 1)$ and $P_i(t)$) as

$$r_i(t) = \ln P_i(t + 1) - \ln P_i(t) \quad (1)$$

where $r_i(t)$ has 1018 observation.

Before establishing the portfolio selection process, we compute the mean return and standard deviation of each stock i .

3. Theoretical background and methodology

We first calculate the price change of the stocks $i = 1, \dots, N$, over a time scale Δt (otherwise known as returns) in order to quantify correlations, see (Plerou et al 2001) and (Sharifi et al. 2004). Denote the price of i^{th} asset at time t as $S_i(t)$, and define its price change as

$$G_i(t) = \ln S_i(t + \Delta t) - \ln S_i(t)$$

Since there is variation in the levels of volatility (standard deviation) of different stocks, we use returns that are normalized with respect to standard deviations σ_i as follows:

$$g_i(t) = \frac{G_i(t) - \langle G_i \rangle}{\sigma_i} \quad (2)$$

where $\sigma_i = \sqrt{\langle G_i^2 \rangle - \langle G_i \rangle^2}$ is the standard deviation of G_i for the assets $i = 1, \dots, N$ and $\langle \dots \rangle$ is denoted as the time average of G_i over the period studied which can be computed as follows, $\langle G_i \rangle = \frac{1}{L} \sum_{l=0}^{L-1} G_i$.

In this paper, our $N = 82$ and $L = 1018$, where N and L correspond respectively to the numbers of the stocks listed in NSE and the days studied. Although NSE has about 188 stocks listed, so many stocks were not consistently traded over the study period. Hence, we choose 82 stocks that were consistent in the market during the period studied. We then compute the equal-time cross-correlation matrix C with elements:

$$C_{ij} \equiv \langle g_i(t), g_j(t) \rangle \quad (3)$$

Being correlation coefficients, the elements of C_{ij} are restricted to domain $-1 \leq C_{ij} \leq 1$ during construction of the C_{ij} , where $C_{ij} = 1$ corresponds to perfect positive correlation, $C_{ij} = 0$ corresponds to uncorrelated pairs of stocks and $C_{ij} = -1$ corresponds to perfect negative correlation.

(Plerou et al. 2001, 2000) and (El Alaoui 2015) note that two main difficulties arise in the analysis of the correlations between any two stocks or more generally, the correlation structure of a portfolio of assets in a financial market. Firstly, market conditions change with time. Hence, the correlation C_{ij} between any two pairs i, j of the stocks may not be stationary. Secondly, time averaging over a finite time series introduces ‘measurement noise’. These facts have implications for portfolio selection and optimisation, given the centrality of cross-correlations among assets to Markowitz portfolio optimization. Recall that the basic tenets of Markowitz’s theory of optimal portfolios are a) to determine the optimal weights of assets with given average returns and risks which maximizes the overall returns for a fixed level of risk, or b) minimises the risk for a given level of overall return. For this purpose, if $\{R_i\}$ are expected returns of the portfolio assets, $\{p_i\}$ are the relative amounts of capital invested in the assets, and $C = (C_{ij})$ is the matrix of covariances of asset returns, then Markowitz’s optimization uses Lagrangian multiplier approach to minimize the overall portfolio variance $\sigma_p^2 = \sum_{i,j=1}^N p_i p_j C_{ij}$ for a given value of overall return $r_p = \sum_{i=1}^N p_i R_i$. The results of this scheme are summarised graphically in mean-variance Efficient Frontiers which shows the range of optimal risk-return combinations possible.

The point, therefore, of Random Matrix Theory (RMT) is to compare the structure of the empirical cross-correlations (in effect covariances) among portfolio assets with the behaviour

of a purely random matrix in which the assets are independent. For this, both eigenstructure and time dependence (or stability) of the matrix C are of interest, and constitute what is known in RMT as dynamics of the correlation matrix. The following notes summarise what is known about these dynamics. For a portfolio of N assets, the matrix C has $N(N-1)/2$ entries to be determined from N time series of length L (for example $N = 82$ and $L = 1018$ in this paper). If L is not very large relative to N , we are basically estimating too many model parameters from sparse information, which introduces the above-mentioned ‘measurement noise’ in empirical correlation matrices. This makes the use of such matrices in applications less accurate in portfolio optimization than alternative matrices filtered for pertinent information using key results of RMT summarised below. Indeed, in RMT this information is gleaned from the behaviour of eigenvalues and eigenvectors of C , compared to a ‘null hypothesis’ purely random matrix such as represented by a finite time series of strictly independent and uncorrelated assets.

In matrix notation, the correlation matrix based on equation (2) above can be expressed as

$$C = \frac{1}{L} G G^T \quad (4)$$

where G is an $N \times L$ matrix with elements $\{g_{im} \equiv g_i(m\Delta t): i = 1, \dots, N; m = 0, \dots, L-1\}$, and G^T is the transpose of G . The RMT method used here is to compare the empirical cross-correlation matrix C against the null hypothesis of a random matrix of the same type. Hence we consider a random correlation matrix

$$R = \frac{1}{L} A A^T \quad (5)$$

where A is an $N \times L$ matrix containing N time series of L random element a_{im} , with zero mean and unit variance that are mutually uncorrelated. (Muirhead 1982) refers to R as Wishart matrices. Deviations from the random matrix results determine the location of pertinent information among set financial assets. By diagonalizing matrix C , we obtain

$$C u_k = \lambda_k u_k \quad (6)$$

where λ_k are the eigenvalues and u_k the eigenvectors, and $k = 1, \dots, N$ is arranged in order of increasing eigenvalues. Statistical properties of random matrices such as R are known; see (Dyson 1971) and (Sengupta & Mitra 1999). In particular, in the limit $N \rightarrow \infty, L \rightarrow \infty$ such that $Q \equiv \frac{L}{N} (> 1)$ is fixed, (Sengupta & Mitra 1999) show analytically that the probability density function $P_{rm}(\lambda)$ of eigenvalues λ of the random correlation matrix R is given by

$$P_{rm}(\lambda) = \frac{Q}{2\pi\sigma^2} \frac{\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}}{\lambda} \quad (7)$$

for λ within the bounds $\lambda_- \leq \lambda_i \leq \lambda_+$, where λ_- and λ_+ are the minimum and maximum eigenvalues of R , respectively, given by

$$\lambda_{\pm} = \sigma^2 \left(1 + \frac{1}{Q} \pm 2\sqrt{\frac{1}{Q}} \right) \quad (8)$$

where σ^2 is equal to the variance of the elements R , Sengupta & Mitra (1999). The eigenvalues of R falls within the interval $[\lambda_-, \lambda_+]$ as predicted by Random matrix theory, Rosenow et al. (2003). σ^2 is equal to unity in the case of a normalised matrix A . Also, the maximum and minimum (theoretical) eigenvalues determine the (theoretical) bounds of the eigenvalues distribution and if some eigenvalues of the matrix C are beyond these bounds, they are said to deviate from the random bound which suggests that they contain pertinent investment information. Being able to detect assets with such real information as opposed to random market noise will enable investors to include them in portfolio constructions for more optimal risk control.

It is also known that the first three eigenvalues represent the overall market information based on the random behaviour of investment returns (random walk hypothesis), which the random matrix represents. Hence, deviations between realised values of the first three eigenvalues (especially the first one) and the predicted maximum eigenvalue indicate the extent to which a stock market is consistent with the RMT assumptions.

In line with the assumption of pure randomness and independence, the distribution of the components $\{\mathbf{u}_k(l) | l = 1, 2, \dots, N\}$ of an eigenvector $\mathbf{u}_k$ of a random correlation matrix R should obey the standard normal distribution with zero mean and unit variance given by

$$P_R(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) \quad (9)$$

Inverse Participation Ratio (IPR)

The Inverse participation ratio IPR is used to analyse the structure of the eigenvectors whose eigenvalues are lying outside the noisy band of RMT predicted eigenvalues (Plerou et al 1999, 2000). It measures the number of components that participate significantly in each eigenvector (Gurh et al. 1998). It also indicates the degree to which the distribution of eigenvectors of the empirical correlation matrix deviate from RMT results, particularly distinguishing an eigenvector with roughly equal components and another with a small number of large components.

Let u_l^k denote the l^{th} component of the eigenvector u^k . The IPR is defined as

$$I^k = \sum_{l=1}^N (u_l^k)^4, \quad l = 1, 2, \dots, 82 \quad (10)$$

where $N = 82$ is the number of assets.

The IPR can be seen as the reciprocal of the number of eigenvector components significantly different from zero (El Alaoui, 2015). This can be illustrated by two limiting cases: (i) if the components of the eigenvector are identical $u_1^k \equiv 1/\sqrt{N}$ has $I^k = 1/N$, and (ii) a vector with one component $u_i^k \equiv 1$ and the remainder zero has $I^k = 1$. Practically, inverting an observed IPR estimates the number of active elements in the time series of a financial asset, that is, the number of eigenvector components that contribute more than random noise to the portfolio risk-return characteristics.

4. The Analysis of the empirical results

Eigenvalue analysis

We analysed $N = 82$ stocks from NSE which a total of $L = 1019$ daily closing prices. We found out that the theoretical eigenvalue bounds of the correlation matrix (from RMT predicted), are $\lambda_+ = 1.6484$ and $\lambda_- = 0.5128$ as maximum and minimum eigenvalues respectively. The value of our Q is $L/N = 12.4146$. Also, from the results, it is observed that the largest eigenvalue $\lambda_1 = 4.87$ which is 2.95 (approximately 3) times larger than the predicted RMT value above. The corresponding eigenvector is the ‘market’ itself which has approximately equal components on all the 82 stocks. The pure noise RMT hypothesis may therefore not be consistent with the NSE stocks.

Also 6 out of our eigenvalues deviated from the above RMT eigenvalue spectrum, which accounts for 10.98% of the total eigenvalues (see fig.1a and 1b.). Therefore, this suggests the presence of true information about the stock market in about 11% of the selected stocks and purely random information in the remaining 89% associated with the purely random matrix, (Laloux et al. 1999) and (Plerou et al. 1999).

Distribution of eigenvectors component analysis

Using Figures Fig 2a, Fig 2b, Fig 2c, Fig 2d and Fig 2e below we analyse the distribution of the eigenvectors by comparing the distribution of the eigenvector components that are inside the bound of the RMT prediction (i.e. $[\lambda_-, \lambda_+]$) with those outside the bound. The normal distribution of the eigenvectors u_1 of the first eigenvalue (which represents the market on the selected stocks) is clearly right-tail asymmetric with a positive mean. The data are also non-normal and therefore inconsistent with RMT predictions.

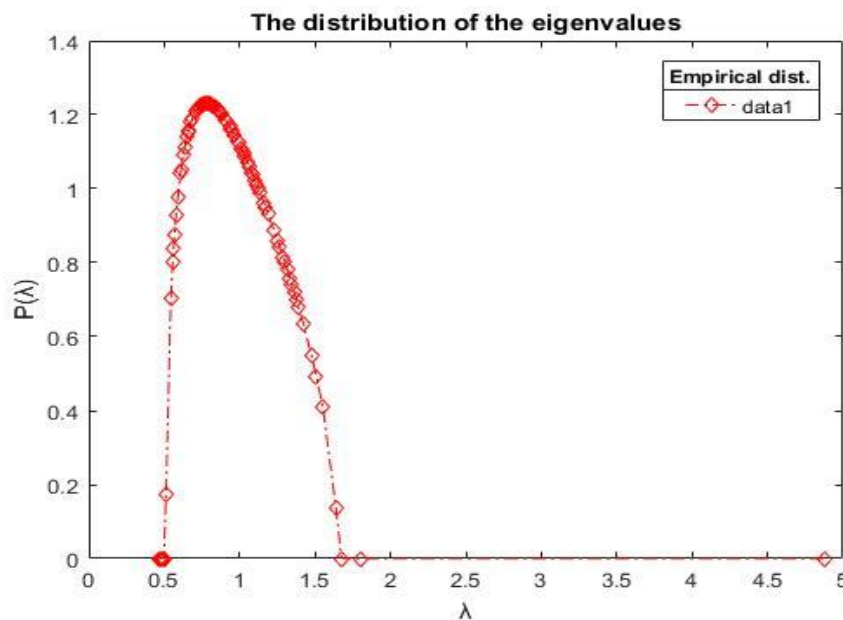


Fig 1a: Empirical distr. of eigenvalues

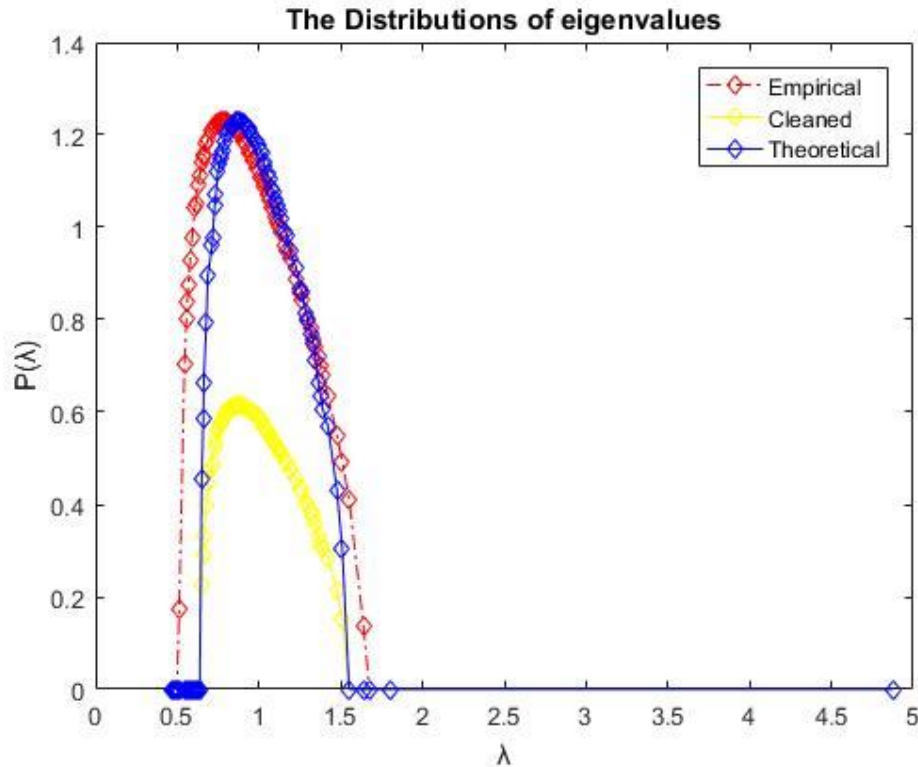


Fig 1b: Theoretical (Marcenko-Pastur), empirical and Cleaned up distr. of eigenvalues

Fig 2b shows a similar non-normal asymmetric distribution of eigenvectors associated with the second eigenvalue, as with the first, but with a reversed left-tailed skew. Normality begins to occur with much higher eigenvalue values such as the eigenvectors associated with the twentieth eigenvalue (see Fig 2c). Thus potentially portfolio-enhancing stocks lay outside but not too far away from predicted RMT the bands. Particularly for the first eigenvalue which represents the market, we can say that the Nigerian Stock Market, as revealed by the 82 stocks, reacts more to positive variations than negative variations.

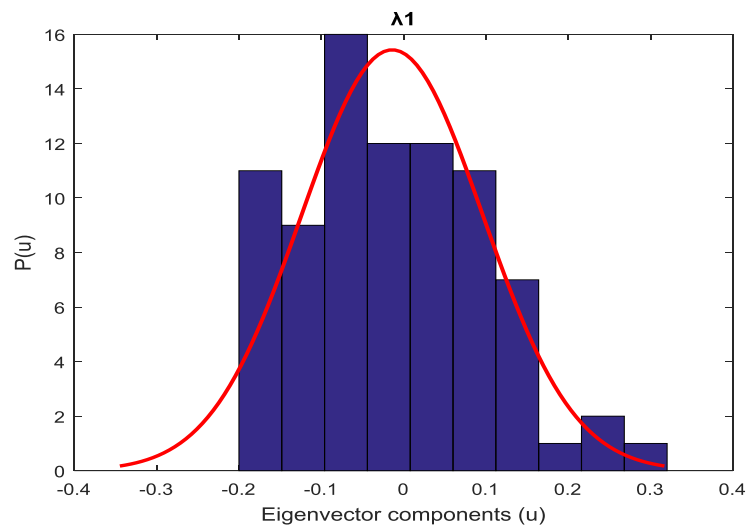


Fig 2a

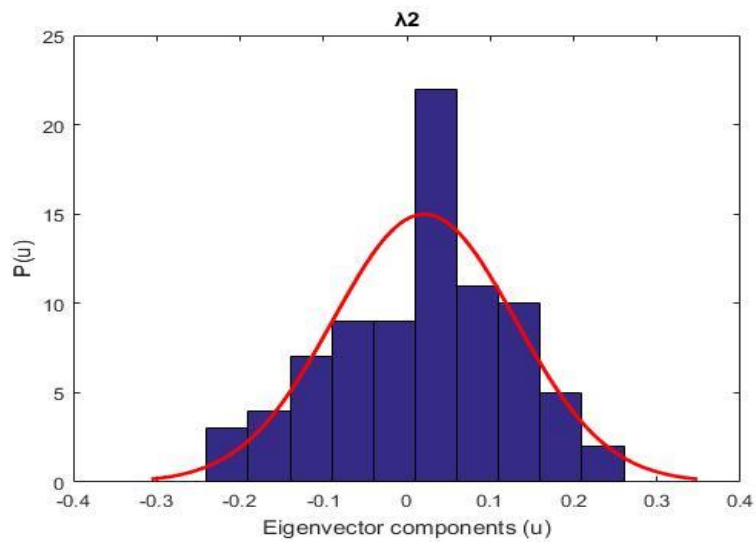


Fig 2b

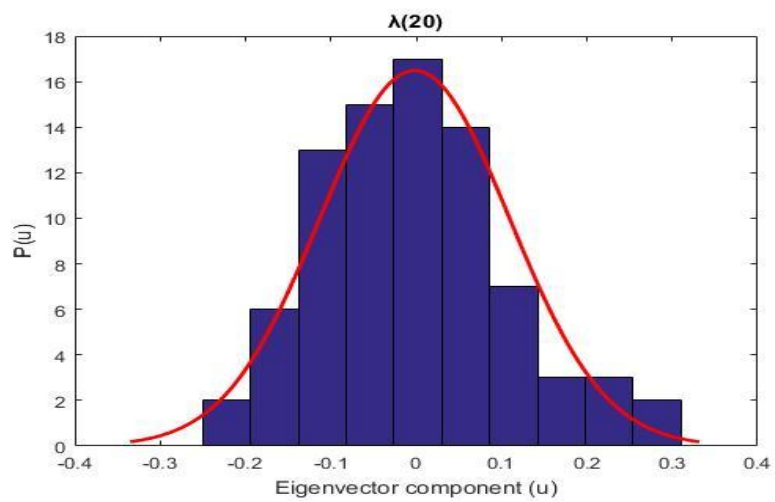


Fig 2c

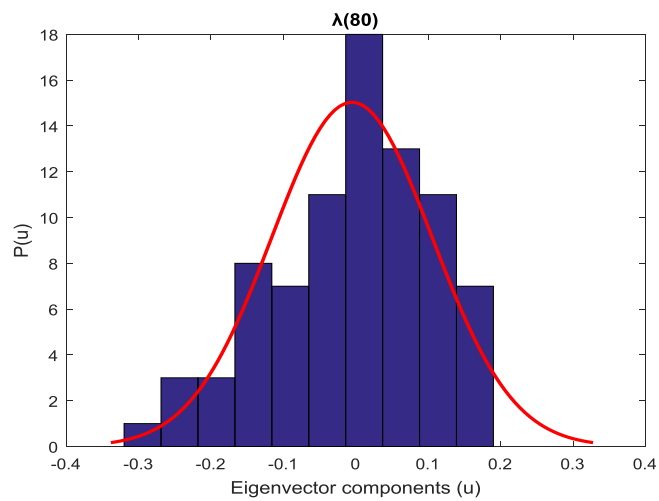


Fig 2d

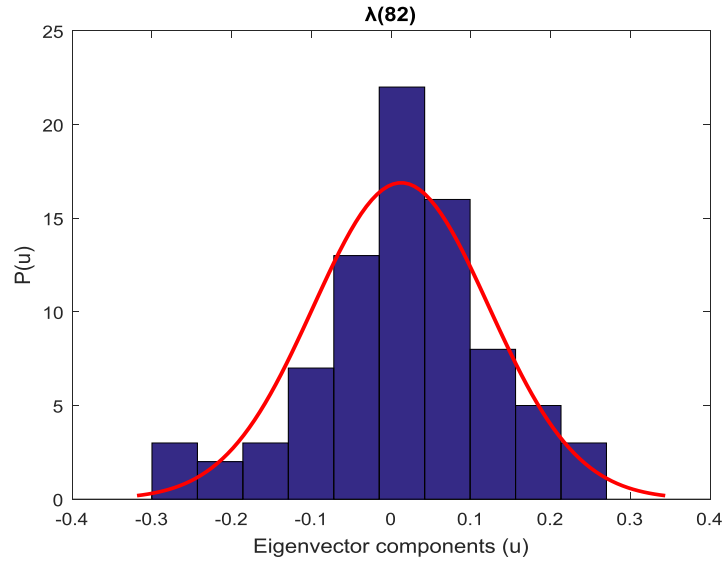


Fig 2e

Inverse participation ratio Analysis

The theoretical mean IPR is approximately equal to $3/N = 3/82 = 0.0366$. A look at Fig. 3 shows that the IPR values are close to a mean level of noise (0.04) indicating that a few stocks do not dominate the dynamics and most stocks participate in the correlation dynamics of asset returns. However, the deviations in IPR values are strong for a few initial elements (17 times more for the highest score 0.17 compared with the lowest non-zero score of about 0.01 between elements 1-6). Overall, there does not appear to be a localisation effect in the spread of IPR values, with very large values for a few elements and very small values for the rest (Alaoui, 2015). It would be interesting to compare this overall market behaviour against sectorial results as mentioned in the introduction to this paper.

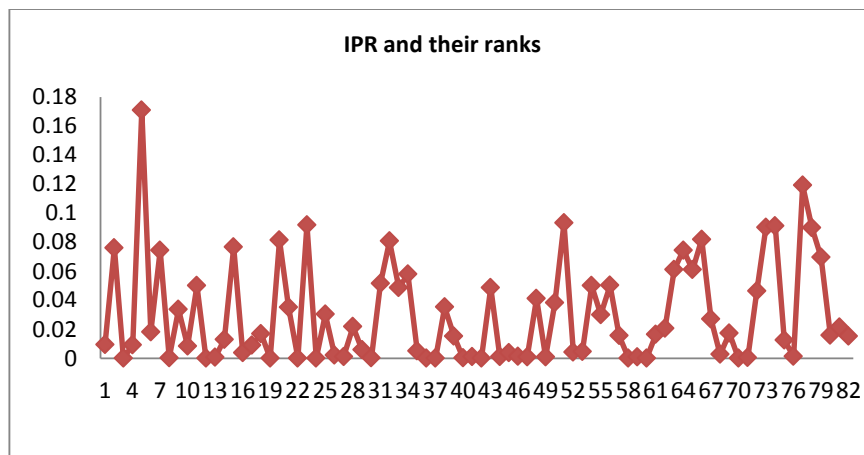


Fig 3: Inverse participation ratio and their rank

Cleaning empirical correlation matrices and stock market characterisation

We noted earlier in the introduction to this paper that estimating empirical correlation matrices introduces measure noise because of the large number of correlation parameters required. Using such matrices further in portfolio optimisation as depicted by Markowitz's mean-variance Efficient Frontiers is therefore subject to inaccuracies in predicting portfolio risks. We examine this in more detail when constructing suitable investment portfolios from overall and sector-based RMT results. For this, we explain briefly how the process works. The first step is to divide the observed series of stock prices and returns into two fairly equal sub-periods. The first period analyses predicted risk and the second realised risk. The second step filters RMT eigenvalues into noisy and non-noisy elements, with the latter situated outside the RMT eigenvalue spectrum. These eigenvalues are maintained and those in RMT bound $[\lambda_-, \lambda_+]$ are cleaned by replacing them with their average values while maintaining the same matrix trace. The third step uses known methods to construct the cleaned correlation matrix from the noisy elements (Alaoui, 2015; Laloux et al. 1999, 2000). After cleaning up the correlation matrix, the return increases and the risk is reduced by 13.7% (see Fig 4 below).

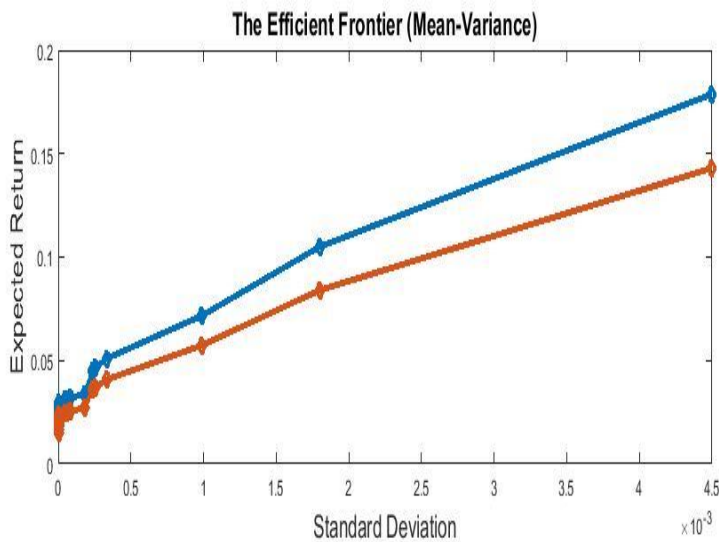


Fig 4

Hints on future work

Stock characterisation using RMT theory will, as noted in the introduction, be used to perform similar analyses as in this paper for different sectors of the Nigerian Stock Market (NSM) and the results compared. Also, portfolio optimisation with cleaned empirical correlation matrices and stocks and bonds selected within and across different market sectors will reveal a menu of potential investment options for market participants. Hence, RMT characterisation should be applied to returns from asset classes and trade-related indices (bond yields, interest dividends, exchange rates, for example). These are potentially interesting directions of future work which this paper engenders in the NSM. The same techniques can also apply to other emerging African markets and combinations of assets across markets.

We also feel that RMT characterisation has potential in measuring extent of closeness or otherwise in selected sectors of these markets, for example the financial services sectors of the NSM and the Johannesburg Stock Exchange (JSE). This benchmarking motive has powerful implications for developing instruments in markets where they are not yet operational (NSM for example) using information from those in which they are traded (the JSE in this case). This line of work is currently in progress by the authors.

5. Conclusion

RMT enabled us to analyse in some detail the correlation structure of stock returns in the Nigerian Stock Exchange. Marcenko-Pastur distribution predicted a theoretical eigenvalue range of between 0.52 and 1.65 approximately. About 6 out of 82 eigenvalues of the selected stocks were outside this eigenvalue spectrum, indicating that about 11% of the stocks have important information that can be used in constructing portfolios with more stable returns and risk characteristics, than a null hypothesis purely random market allows in the remaining 89% cases.

The eigenvectors associated with the most important eigenvalues (λ_1, λ_2) are non-normal and highly asymmetric which is inconsistent with RMT. This fact suggests the presence of market signals such as depicted also in the information-carrying stocks. Given that the first eigenvalue represents the market, it also suggests that the Nigerian Stock Market (NSM) is inefficient, a fact that is consistent with known results in (Ezepue and Omar 2012), for example. Importantly, this positive asymmetry in distribution of eigenvectors of λ_1 shows that the NSM reacts more strongly to positive variations (and news) than negative variations.

Whilst the RMT bounds imply that the majority of stocks analysed (89%) will suitably follow a zero-mean normal distribution in their returns, the normal distribution is not a good enough fit for the materially informative stocks outside the RMT eigenvalue bounds. Finally, the inverse participation ratio gives additional insights about the spread or localisation of eigenvectors by their ranks. It shows a fairly even spread which suggest that there are no dominant stocks in among the 82 stocks.

These insights are useful for constructing more optimal portfolios. For example, RMT eigenstructure results are used to clean the empirical correlation matrices and thereby improve the realised and predicted risks associated with Markowitz mean-variance Efficient Frontier. Also, detailed risk analysis of individual stock and portfolio returns outside RMT bounds should use suitable non-normal distributions.

Future lines of work along these lines also include the development of financial derivatives in the NSM using information on NSM-JSE market affinities which we foreshadowed in this paper.

References

Bouchaud, J.-P. Potters, M. (2003). Theory of Financial Risk and Derivative Pricing: From Statistical. *Physics to Risk Management Cambridge University Press*, Cambridge

Cao, G. Longbing, X. Cao, J. (2012). Multifractal detrended cross-correlations between the Chinese exchange and stock market. *Physica A*, 391, pp. 4855–4866

Cao, G. Cao, J. Xu, L. He, L.-Y. (2014). Detrended cross-correlation analysis approach for assessing asymmetric multifractal detrended cross-correlations and their application to the Chinese financial market. *Physica A*, 393, pp. 460–469

Çukura, S. Eryiğita, M. Eryiğitb, R. (2007). Cross correlations in an emerging market financial data. *Physica A* 376, pp 555 - 564.

Daly, J. Crane, M. Ruskin, H. (2008). Random matrix theory filters in portfolio optimisation: A stability and risk assessment. *Physica A*, 387, pp. 4248–4260

Dyson, F. (1962). Statistical theory of the energy levels of complex systems I, II and III. *J. Math. Phys.*, 3 (1), pp. 140–175

Dyson, F. (1971). Distribution of eigenvalues for a class of real symmetric matrices. *Rev. Mexicana Fís*, 20, pp. 231–237

Dyson, F. Mehta, M. L. (1963). Statistical theory of the energy levels of complex systems IV. *J. Math. Phys.* Vol.4, no.5, 701-712;

El Alaoui, M. Benbachir, S. (2013). Multifractal detrended cross-correlation analysis in the MENA area. *Physica A*, 392, pp. 5985–5993.

El Alaoui, M. (2015). Random matrix theory and portfolio optimization in Moroccan Stock Exchange. *Physica A*, 433, 92-99.

Ezepue, P. O. Omar, M. A. T. (2012). Weak-form market efficiency of the Nigerian Stock Market in the context of financial reforms and global financial crisis. *Journal of African Business*, 13(2), pp. 209-220.

Fyodorov, Y.V. Mirlin, A.D. (1992). Analytical derivation of the scaling law for the inverse participation ratio in quasi-one-dimensional disordered systems. *Phys. Rev. Lett.*, 69, p. 1093

Fyodorov, Y.V. Mirlin, A.D. (1993). Level-to-level fluctuations of the inverse participation ratio in finite 1D disordered systems. *Phys. Rev. Lett.*, 71, p. 412

Fyodorov, Y.V. Mirlin, A.D. (1994). Statistical properties of eigenfunctions of random quasi 1D one-particle Hamiltonians. *Internat. J. Modern Phys. B*, 8, p. 3795

Guhr, T. Müller-Groeling, A. Weidenmüller, H. (1998). Random matrix theory in quantum physics. *Phys. Rep.*, 299, pp. 198–425

He, L.-Y. Chen, S.-P. (2011). A new approach to quantify power-law cross-correlation and its application to commodity markets. *Physica A*, 390, pp. 3806–3814

He, L.-Y. Chen, S.-P. (2011). Nonlinear bivariate dependency of price–volume relationships in agricultural commodity futures markets: A perspective from multifractal detrended cross-correlation analysis. *Physica A*, 390, pp. 297–308

Izenman, A. J. Introduction to Random matrix theory.

<http://astro.ocis.temple.edu/~alan/MMST/IntroRMT.PDF>

Kulkarni, V. Deo, N. (2007). Correlation and volatility in an Indian stock market: A random matrix approach. *Eur. Phys. J. B* 60: 101. doi:10.1140/epjb/e2007-00322-1

Laloux, L. Cizeau, P. Bouchaud, J.-P. Potters, M. (1999). Noise dressing of financial correlation matrices. *Phys. Rev. Lett.*, 83, pp. 1467–1470

Laloux, L. Cizeau, P. Potters, M. Bouchaud, J.-P. (2000). Random matrix theory and financial correlations. *Int. J. Theor. Appl. Finance*, 3, pp. 1–6

Lee, P. Ramakrishnan, T. (1985). Disordered electronic systems. *Rev. Modern Phys.*, 57, p. 287

Markowitz, H. (1952). Portfolio selection. *J. Finance*, 7, pp. 77–91

Mehta, M. L. (2004). Random Matrices. 3rd Edition, *Pure and applied mathematics (Amsterdam)*, 142, Amsterdam, Netherlands: Elsevier/Academic Press.

Mehta, M. L. (1991) "Random Matrices." *Spectral statistics in elastodynamics* (1881).

Mirlin, A. Fyodorov, Y. (1993). The statistics of eigenvector components of random band matrices: analytical results. *J. Phys. A: Math. Gen.*, 26, p. 551

Mu, G.-H. Zhou, W.-X. Chen, W. Kertész, J. (2010). Long-term correlations and multifractality in trading volumes for Chinese stocks. *Physics Procedia*, 3, pp. 1631–1640

Pafka, S. Kondor, I. (2003). Noisy covariance matrices and portfolio optimization II. *Physica A*, 319, pp. 487–494

Pafka, S. Kondor, I. (2004). Estimated correlation matrices and portfolio optimization. *Physica A*, 343, pp. 623–634

Pan, R.K. Sinha, S. (2007). Collective behaviour of stock price movements in an emerging market. *Phys. Rev. E* 76, 046116

Plerou, V. Gopikrishnan, P. Rosenow, B. Amaral, L.A. Stanley, H.E. (1999). Universal and non-universal properties of cross correlations in financial time series. *Phys. Rev. Lett.*, 83, pp. 1471–1474.

Plerou, V. Gopikrishnan, P. Rosenow, B. Amaral, L.A. Stanley, H.E. (2000). A random matrix theory approach to financial cross-correlations. *Physica A*, 287, 374–382.

Plerou, V. Gopikrishnan, P. Rosenow, B. Amaral, L.A. Stanley, H.E. (2001). Collective behaviour of stock price movements: A random matrix theory approach. *Physica A*, 299, 175–180.

- Plerou, V. Gopikrishnan, P. Rosenow, B. Amaral, L.A. Guhr, T. Stanley, H.E. (2002). Random matrix approach to cross correlations in financial data. *Phys. Rev. E*, 65, p. 066126
- Podobnik, B. Stanley, H. (2008). Detrended cross-correlation analysis: A new method for analyzing two nonstationary time series. *Phys. Rev. Lett.*, 100, p. 084102
- Podobnik, B. Horvatic, D. Petersen, A.M. Stanley, H.E. (2009). Cross-correlations between volume change and price change. *Proc. Natl. Acad. Sci. USA*, 106, pp. 22079–22084
- Rak, R. Drozd, S. Kwapien, J. (2007). Nonextensive statistical features of the Polish stock market fluctuations. *Physica A* 374, pp 315–324
- Rak, R. Kwapie, J. Drozd, S. Oswiecimka, P. (09/2008). Cross-correlations in Warsaw Stock Exchange. *Acta Physica Polonica A*, Vol.114 (3), pp.561-568
- Sengupta, A.M. Mitra, P.P. (1999). Distributions of singular values for some random matrices. *Phys. Rev. E*, 60, pp. 3389–3392.
- Sharifi, S. Crane, M. Shamaie, A. Ruskin, H. (2004). Random matrix theory for portfolio optimisation: a stability approach. *Physica A*, 335, 629-643.
- Wang, D.-H. Suo, Y.-Y. Yu, X.-W. Lei, M. (2013). Price-volume cross-correlation analysis of CSI300 index futures. *Physica A*, 392, pp. 1172–1179
- Wang, G.-J. Xie, C. He, L.-Y. Chen, S. (2014). Detrended minimum-variance hedge ratio: A new method for hedge ratio at different time scales. *Physica A*, 405, pp. 70–79
- Wang, G.-J. Xie, C. Chen, S. Yang, J.-J. Yang, M.-Y. (2013). Random matrix theory analysis of cross-correlations in the US stock market: Evidence from Pearson's correlation coefficient and detrended cross-correlation coefficient. *Physica A*, 392, pp. 3715–3730
- Wigner, E. (1951). On a class of analytic functions from the quantum theory of collisions. *Ann. of Math.*, 53, pp. 36–67
- Wigner, E.P. (1955). Characteristic vectors of bordered matrices with infinite dimensions. *Ann. of Math.*, 62, pp. 548–564.
- Wilcox D. Gebbie, T. (2007). An Analysis of cross-correlations in an emerging market. *Physica A*, 375, pp584-598.
- Zebende, G.F. (2011). DCCA cross-correlation coefficient: Quantifying level of cross-correlation. *Physica A*, 390, p. 614
- Zhou, W.-X. (2008). Multifractal detrended cross-correlation analysis for two nonstationary signals. *Phys. Rev. E*, 77 (6), p. 066211